1 Solution — GIAM 5.1 Problem 2

1.1 Problem

What is wrong with the following inductive proof that “all horses are the same color”?

Theorem. Let H be a set of n horses. All horses in H are the same color.

Proof. By induction on n.

Basis. If $|H| = 1$, the single horse is the same color as itself.

Inductive step. Given a set of $k+1$ horses

$$H = \{h_1, h_2, \ldots, h_{k+1}\},$$

define

$$H_a = \{h_1, h_2, \ldots, h_k\}, \quad H_b = \{h_2, h_3, \ldots, h_{k+1}\}.$$

By the inductive hypothesis, all horses in H_a are the same color, and all horses in H_b are the same color. The horses $h_2, \ldots, h_k$ lie in both sets, so every horse in H shares this common color.

1.2 Where the Argument Fails

The inductive step silently assumes that H_a and H_b **share at least one horse**, so that the two monochromatic groups can be linked into a single color.

Consider the very first non-trivial case, $k = 1$ (i.e., going from $n = 1$ to $n = 2$). Let $H = \{h_1, h_2\}$. Then

$$H_a = \{h_1\}, \quad H_b = \{h_2\}, \quad H_a \cap H_b = \varnothing.$$

There is no horse in both sets, so the argument “horses $h_2, \ldots, h_k$ lie in both sets” is vacuous — it refers to an empty range of indices. Consequently, nothing forces h_1 and h_2 to have the same color, and the implication $P(1) \Rightarrow P(2)$ does not hold.

1.3 Conclusion

The induction breaks at the step $\boldsymbol{k = 1 \to k+1 = 2}$, because the two $\boldsymbol{k}$-element subsets used to “bridge” the colors are disjoint. Since this single link in the inductive chain fails, the entire proof is invalid, and indeed the theorem itself is false.